Manikanta Cheruku

+91 8074695041 | cherukumanikanta77@gmail.com | linkedin.com/in/manikanta-cheruku
github.com/manikanta7cheruku | https://manikantacheruku.vercel.app | Hyderabad, India

EDUCATION

Sree Dattha Institute of Engineering and Science Hyderabad, India
B.Tech Computer Science and Business Systems 2023 - 2027
CGPA: 7.0/10 | Self-Study: System Design, Vector Databases, LLM Integration

SKILLS

Languages: Python, JavaScript, Java, SQL, TypeScript
AI/ML: Ollama, Whisper STT, ChromaDB, SentenceTransformers, LangChain, XGBoost, SHAP, RAG, Vector Embeddings, Agentic AI, Local LLM Inference
Backend: FastAPI, REST API Design, Async Programming, Node.js
Frontend: React.js, Electron, Vite, Tailwind CSS, HashRouter
Databases: PostgreSQL, SQLite, ChromaDB (Vector Database)
Cloud & DevOps: Git, GitHub, Docker, Linux, Render, CI/CD
Methodologies: Agile, OOP, REST API Design, System Design, TDD Basics

PROJECTS

SEVEN [Live] 2025 - Present
Python, FastAPI, Electron, React, Whisper, Ollama, ChromaDB

- Developed a **fully offline AI voice assistant** for Windows integrating **4 AI components** (Whisper STT, Ollama LLM, ChromaDB vector DB, SentenceTransformers) with 100% zero cloud dependency, eliminating recurring API costs and ensuring **full user data privacy.**
- Architected a local **RAG memory system** with ChromaDB enabling semantic search across conversation history. Sub-second vector retrieval on consumer hardware (8GB RAM, no GPU required) .
- Implemented **30+ FastAPI endpoints** orchestrating voice pipeline, license validation, telemetry sync, and IPC between Electron shell and Python backend with PostgreSQL on Render.
- **Shipped solo as complete installable Windows product with under 5 minute setup time product.** Multi-step onboarding wizard, tier-based licensing, referral system. Released publicly on GitHub.

AgentFetch [Live]
Python, FastAPI, React, REST APIs, CLI

- **Crafted** agentic data pipeline integrating **2 live APIs** (crypto + weather) with async FastAPI backend, React dashboard, and developer CLI.
- **Designed extensible agent interface layer** supporting future LLM orchestration with **zero architectural refactoring** required

Showtime-Sentinel [Github]
Python, asyncio, SQLite, Playwright, Telegram Bot, pytest

- Built an **async** movie-ticket availability monitor with **SQLite** persistence, Playwright automation, CLI, and Telegram notifications.
- Implemented state-based change detection, deduplication, retries, and **117 automated tests** covering core workflows and database persistence.

EXPERIENCE

Solo Developer | SEVEN (Independent Product) Jan 2026 - Present
Python, FastAPI, Electron, React, Whisper, Ollama, ChromaDB

- *Gained expertise in **end-to-end product development expertise** from system design through **deployment and ongoing maintenance.***
- *Mastered **AI integration patterns** combining speech recognition, language models, and vector databases into unified production pipelines.*

LANGUAGES

English (Fluent) · Hindi (Fluent) · Telugu (Native) · Spanish (Basic)